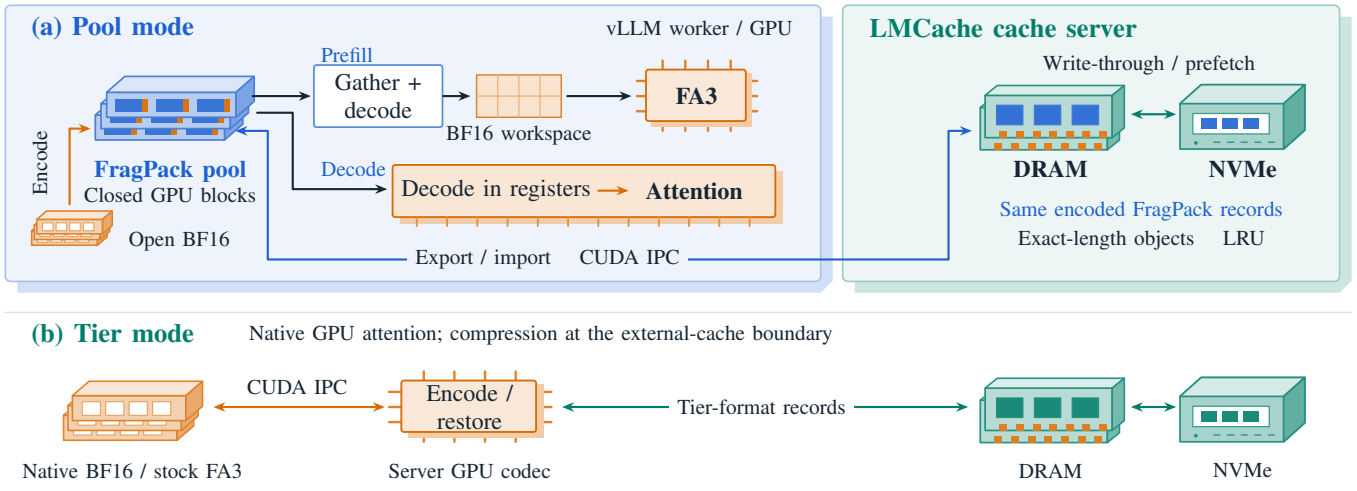
(a) Pool mode
vLLM worker / GPU
Prefill
Gather + decode
BF16 workspace
FA3
Encode
FragPack pool
Closed GPU blocks
Open BF16
Decode
Decode in registers → Attention
Export / import
CUDA IPC
LMCache cache server
Write-through / prefetch
DRAM
NVMe
Same encoded FragPack records
Exact-length objects
LRU
(b) Tier mode
Native GPU attention; compression at the external-cache boundary
CUDA IPC
Encode / restore
Tier-format records
Native BF16 / stock FA3
Server GPU codec
DRAM
NVMe